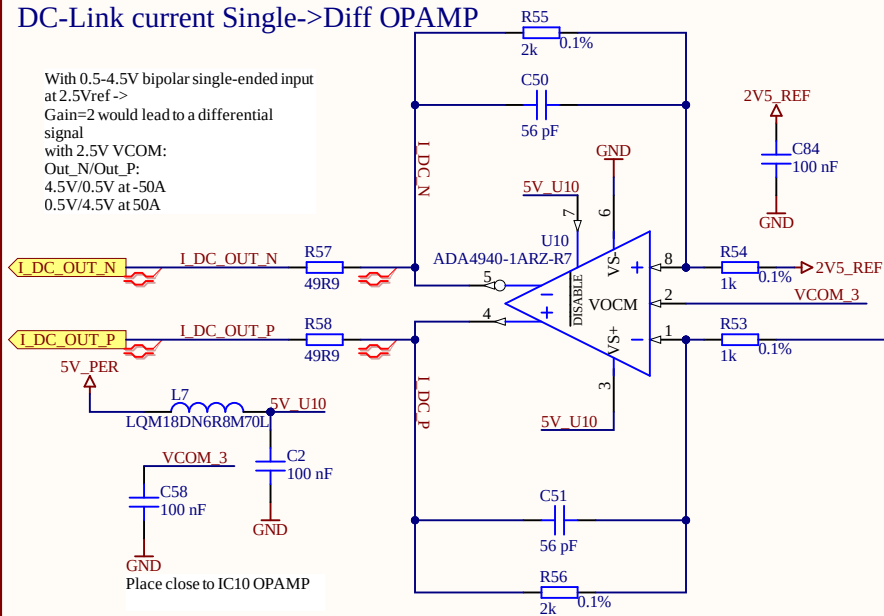


DC-Link current Single->Diff OPAMP

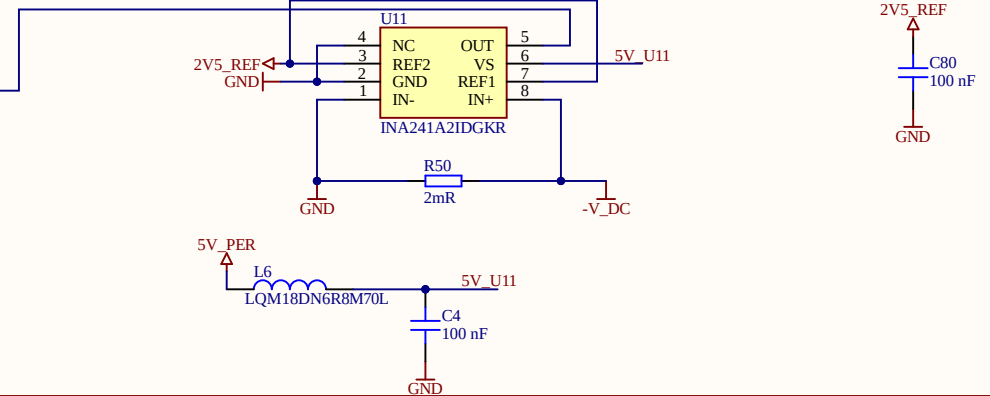
With 0.5-4.5V bipolar single-ended input at 2.5Vref ->
Gain=2 would lead to a differential signal with 2.5V VCOM:
Out_N/Out_P:
4.5V/0.5V at -50A
0.5V/4.5V at 50A



50A max @2mOhm =
1000mV
With fixed 20V/V Gain and
2.5V ref ->
0.5- 4.5V bipolar
single-ended

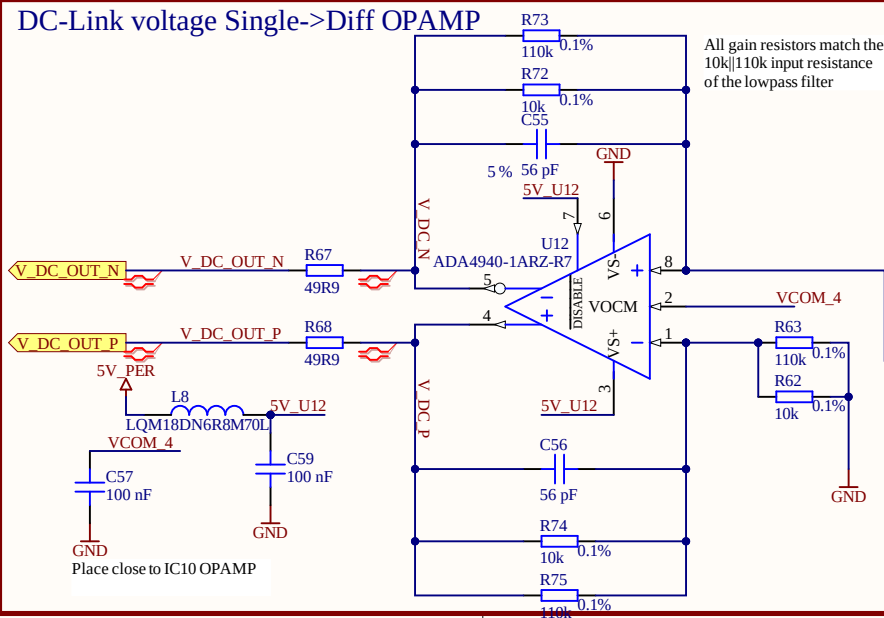
U11 the terminals IN- and IN+ are connected inverted. This is done to improve the PCB-layout. This is fixed, by inverting the inputs at U10 again.

DC-Link current measurement



DC-Link voltage Single->Diff OPAMP

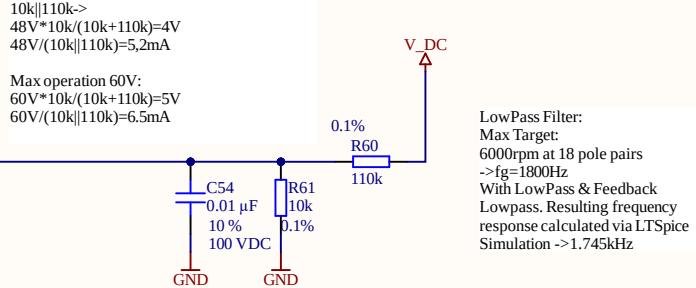
All gain resistors match the 10k||110k input resistance of the lowpass filter



Voltage measurement scaled with voltage divider R60 and R61 :
Max 48V (LowPass max 60V as buffer)
10k||110k->
 $48V \cdot 10k / (10k + 110k) = 4V$
 $48V / (10k || 110k) = 5.2mA$

Max operation 60V:
 $60V \cdot 10k / (10k + 110k) = 5V$
 $60V / (10k || 110k) = 6.5mA$

DC-Link Voltage lowpass



LowPass Filter:
Max Target:
6000rpm at 18 pole pairs
->fg=1800Hz
With LowPass & Feedback
Lowpass. Resulting frequency
response calculated via LTSpice
Simulation ->1.745kHz

Title DC-link.SchDoc

Revision: Rev04

Design Engineer: K.Patel/D. Hufnagel

Project: UZ_D_Inverter.PrjPCB

UltraZohm

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Date: 1/3/2025

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